

Milestone 1 - Fabrication and Development Environment

AR4 MK4 Robotic Arm Project

Team name: Team 2

Team members: Brandon Newton, Andrew Overton, Roshan Padhan

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Repository: <https://github.com/BengalPirate/Fabrication-and-Development-Environment>

1. Team and Preliminary Responsibilities

Member	Primary responsibility	Also supporting
Brandon Newton	Embedded programming (firmware, ESP32/controller, serial comms)	Sensors and computer vision; documentation and repository
Andrew Overton	Mechanical assembly and fabrication (3D printing, assembly, tolerancing)	Bill of materials
Roshan Padhan	Electronics and wiring (drivers, motors, power, harness)	Procurement

Responsibilities may be shared or reassigned as the project progresses.

2. Development Environment - Installation and Verification

All required tools were installed and verified (see the software-setup summary in the repository, docs/software-setup-summary.md). Verification screenshots are in Appendix A.

Tool	Installed	Verified by	Screenshot
Visual Studio Code	Yes	Opens with the project folder	Fig. A-1
Python 3	Yes	<code>python --version -> Python 3.13.9</code>	Fig. A-2
Git	Yes	<code>git --version -> 2.50.1 (Apple Git-155)</code> ; repo cloned	Fig. A-2
Arduino IDE + ESP32 board support/driver	Yes	ESP32 board + /dev/cu.usbserial-10 port selectable; sketch compiles and uploads	Fig. A-4
Slicer / fab-file software (Bambu Studio)	Yes	AR4 STL parts open and slice	Fig. A-3
AR4 control software (Ver 6.4.1)	Yes	GUI launches on the PC (Teensy not yet connected)	Fig. A-6

Tool	Installed	Verified by	Screenshot
AR4 project files	Yes	Repository + AR4 STLs open (VS Code / Bambu Studio)	Figs. A-1, A-3

The only notable issue was an ESP32 **upload failure during the high-speed flash handshake** (esptool reported *"Unable to verify flash chip connection"* when switching to 921600 baud). It was resolved by lowering the Arduino IDE upload speed and using a data-capable USB cable connected directly to the computer; the sketch then compiled and uploaded successfully (Fig. A-4).

3. ESP32 Verification Test

Program: `firmware/esp32_verification/esp32_verification.ino` (in the repository).

The program:

- **Compiles and uploads** successfully to the ESP32 (chip ESP32-D0WD-V3, rev v3.1) over `/dev/cu.usbserial-10`, using the Arduino IDE ESP32 core.
- Produces a **visible output**, the on-board LED blinks at 1 Hz.
- **Prints to the Serial Monitor**, a heartbeat message twice per second at 115200 baud.
- Includes **header comments** identifying the team and **section comments** explaining configuration, `setup()`, and `loop()`.

Results:

- Compilation and upload: successful. esptool reported *"Wrote 285280 bytes ... Hash of data verified. Hard resetting via RTS pin..."* and the board reset and ran the sketch (screenshot Fig. A-4).
- Serial Monitor output at 115200 baud (screenshot Fig. A-5):

```
=====
AR4 Project - ESP32 Verification (Milestone 1)
Team members: Brandon Newton, Andrew Overton, Roshan Padhan
Status: ESP32 online. Beginning blink + serial test...
=====
Heartbeat 0 -> LED ON
Heartbeat 0 -> LED OFF
Heartbeat 1 -> LED ON
...
```

4. AR4 Fabrication and Component Review

(This section is generated from the fabrication checklist, docs/ar4-fabrication-checklist.md. See that file for the full component table.)

4.1 Major mechanical components

The AR4's load-bearing structure is **machined aluminum**, not printed.

Item	Detail	Source
Aluminum parts kit	26 machined aluminum pieces , the links/housings for J1-J6	Purchased (Annin "Aluminum Parts Kit")

Item	Detail	Source
Joint bearings	Angular-contact / taper / thrust / needle bearings at each joint (see Section 4.3)	Purchased (hardware kit)
Timing belts and pulleys	Belt-driven joints (MK3/4 use HTD-series belts on the belt-driven axes)	Purchased (hardware kit)
Fasteners and shafts	Metric machine/set screws, dowel pins, precision shafts	Purchased (hardware kit)
End effector (optional)	Servo gripper or pneumatic gripper	Purchased (optional kit)

4.2 Components that must be 3D printed

Printed parts are **covers, enclosure panels, spacers, and switch/sensor mounts**, secondary/protective, not the primary structure.

Item	Notes
Base enclosure / panels	Houses electronics in the base
Guards / covers	Protective covers over joints
Homing-switch and sensor mounts	Mounts for limit switches
Spacers / small brackets	Misc. printed hardware

- **File format: STL** (free download from Annin's downloads page / GrabCAD / Thingiverse).
- **Material: PETG** recommended (Annin's own printed sets are PETG).
- **Print settings (Annin spec):** approximately 5 perimeters/shells, 25% infill, supports where overhangs require.
- **Filament needed:** roughly 1 spool for a full set; a pre-printed set can be purchased instead.
- **Note:** exact printed-part count is approximate, confirm from the STL package.

4.3 Purchased mechanical and electronic components

Bearings: 30203 taper roller (17x40x13.25 mm), 35x52x4 thrust, HK1612 needle roller (16x22x12 mm), B1616 needle roller, TRD1625 thrust washers, LM3UU linear bushings.

Motion hardware: HTD timing belts + pulleys on the belt-driven joints (note: verify exact belt pitch per joint against the MK4 manual), precision shafts (e.g., 3 mm x 85 mm), dowel pins, metric fasteners / set screws. Reduction is via motor planetary gearboxes + timing belts, **no harmonic or cycloidal drives**.

4.4 Motors, drivers, sensors, control hardware

Motors: geared NEMA steppers with integrated magnetic encoders (closed-loop), from StepperOnline:

Joint	Motor (MK3/4-era SKU)	NEMA size	Gear ratio
J1	17HS15-1684D-EG10-AR4	NEMA 17	10:1
J2	23HS22-2804D-YGS50-AR4	NEMA 23	50:1
J3	17HS15-1684D-EG50-AR4	NEMA 17	50:1
J4	11HS20-0674D-EGS16-AR4	NEMA 11	16:1
J5	17LS19-1684E-200G-AR4 (pancake)	NEMA 17	belt / ~200-step
J6	14HS11-1004D-EGS20-AR4	NEMA 14	20:1

Drivers (6, one per joint): StepperOnline/Leadshine digital: **DM332T x3** (J1-J3), **DM320T x3** (J4-J6). Note: DIP-switch (microstepping) settings must match the firmware/software version.

Controller / electronics: Teensy 4.1 (600 MHz Cortex-M7) main motion controller; Arduino Nano for 5 V peripherals (gripper/solenoid, inside the base enclosure in MK4); ACS712 current sensor for gripper over-current protection; 2x12 terminal board; USB serial to PC.

Sensors: integrated magnetic encoder on each motor (closed-loop feedback to Teensy); mechanical limit switches for homing per axis (SV-166-1C25; KW12 for J5).

Power: 24 VDC supply, StepperOnline LYD2409000 (24 V, ~9 A, approximately 216 W); robot max draw ~198 W. A pneumatic gripper (if used) needs a separate 24 VDC brick + air supply.

4.5 Currently unavailable parts / materials

Inventory to be confirmed by the team. Longest-lead-time item is the **geared+encoder motor set, order early**. Also verify availability of the **Teensy 4.1** (industry-wide supply fluctuations). Remaining items (aluminum parts kit, hardware kit, 24 V PSU, ~1 spool PETG, optional gripper, 3D-printer access) to be checked against the team's on-hand inventory.

4.6 Anticipated fabrication or assembly problems

- **J2 backlash**, most-cited accuracy issue; play amplifies down the kinematic chain. Set J2 belt/gear tension carefully.
- **Fragile homing switches**, easy to break a limit switch; follow the homing/config order.
- **Wiring density**, 6 motors + 6 encoders + 6 limit switches + gripper crammed into the base. Continuity-check every wire before power-up.
- **PETG printing**, warping/adhesion and support removal on covers; watch tolerances on brackets around bearings/switches.
- **Belt tensioning**, belt-driven joints need correct tension for low backlash.
- **Mandatory calibration**, required after flashing firmware and after any Teensy power cycle.
- **Driver DIP switches**, must match firmware/software version (MK3+).

4.7 Opened fabrication file

AR4 printable-part **STL** files were opened and sliced in **Bambu Studio** (targeting a Bambu Lab H2C, 0.20 mm layer height), see screenshot Fig. A-3. This confirms the team can open and prepare the AR4 fabrication files for printing.

5. Repository

The shared repository is organized as described in the README and contains the ESP32 code, software-setup summary, fabrication checklist, responsibilities, and a folder structure for future code and documentation. The instructor has been added as a collaborator.

Repository: <https://github.com/BengalPirate/Fabrication-and-Development-Environment>

6. Summary / Readiness

The development environment is fully installed and verified (VS Code, Python 3.13.9, Git 2.50.1, the Arduino IDE with ESP32 support, and Bambu Studio for the fabrication files). The ESP32 board test compiles, uploads, blinks the on-board LED, and streams a heartbeat over serial, confirming the toolchain and PC-to-board communication. The AR4 MK4 bill of materials, printed parts, and control hardware have been reviewed and an AR4 STL was successfully opened and sliced. Next steps are to complete the parts inventory, order the long-lead motor set, and begin authorized fabrication.

Appendix A - Screenshots

[Insert screenshot: figA9-vscode.png]

Fig. A-1: Visual Studio Code open with the project folder and Milestone 1 report.

[Insert screenshot: figA10-python_and_git.png]

Fig. A-2: Python and Git versions verified in the terminal (python --version -> Python 3.13.9; git --version -> 2.50.1).

[Insert screenshot: figA11-slicer.png]

Fig. A-3: AR4 STL parts opened and sliced in Bambu Studio (fabrication file).

[Insert screenshot: figA7-upload.png]

Fig. A-4: ESP32 verification sketch compiled and uploaded successfully in the Arduino IDE.

[Insert screenshot: figA8-serial.png]

Fig. A-5: Serial Monitor output at 115200 baud (LED-blink heartbeat).

[Insert screenshot: figA12-interface.png]

Fig. A-6: AR4 control software (Ver 6.4.1) installed and launched on the PC. The "Unable to establish communications with Teensy 4.1" notice is expected, the controller is not yet connected at this stage.